

Mid Semester Test – III (Nov-2023)

B. E. II Year (IT-A, IT-B)

Subject – Computer Organization and Architecture (ITR3G1)

Duration: 70 Min

Note: Attempt all questions and answer should be to the point.

Max Marks: 20

Q.1	For a given CPU, the first machine cycle M1 always consist of 4-states, and the remaining machine cycles consist either of 3 or 4 states. An instruction cycle consists of a minimum of 2 and a maximum of 5 machine cycles. If the CPU's clock rate is 50 MHz, calculate : The shortest instruction execution time, The longest instruction execution time, If every memory access introduces one wait state.	5	
Q.2	Design a firmware architecture for a digital component that have four states, three inputs (J, K and Clock) and two outputs (ZERO and ONE).	5	
Q.3	Give the principle of linear pipelining with the basic structure of a linear pipeline processor and the space time diagram.	5	
Q.4	Derive the <i>Speedup</i> , <i>Efficiency</i> and <i>Throughput</i> for a k-stage linear pipeline processor.	5	

E-301

22